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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
First Semester B.Tech (CBCS) Degree Examinations
December 2024

ME1105-1 – INTRODUCTION TO INTERNET OF THINGS
(For ME)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five** full questions choosing **Two** full questions from **Unit – I & Unit – II** each and **One** full question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS**20 Marks**

1. What is the process of energy conversion from one form to another, which is fundamental to sensing and actuation in IoT?
A) Conduction
B) Transduction
C) Modulation
D) Amplification
2. Which of the following is NOT an example of human-generated unstructured data mentioned in the content?
A) Text
B) Videos
C) Sensor data
D) Emails
3. In which types of networks are point-to-multipoint connections commonly used?
A) Wired networks and satellite communication
B) Fiber-optic networks and mesh networks
C) Wireless networks and IP telephony
D) Point-to-point networks and LANs
4. In the outlined actuation system, what is primarily responsible for converting human commands into sequential machine-language command sequences?
A) The robotic arm
B) The motor
C) The remote user
D) The processor
5. What is the primary benefit of virtualization in the context of IoT and cloud computing?
A) Reducing the number of interconnected devices
B) Increasing user awareness of data storage and processing locations
C) Simplifying data management for end users
D) Minimizing the risk of system crashes
6. Why is the selection of an appropriate machine learning model considering a challenge?
A) Because it is unnecessary to choose a model in machine learning
B) Because there is only one model that works for all machine learning tasks
C) Because the quality of a model has no impact on the success of an ML-based system
D) Because multiple models may be suitable, and choosing the best-performing one is essential
7. What are the two main types of nodes in a decision tree?
A) Start and Stop nodes
B) Input and Output nodes
C) Decision and Leaf nodes
D) Root and Branch nodes
8. What is the primary objective of the k-means clustering algorithm?
A) To classify data into predefined categories
B) To reduce the dimensionality of the dataset
C) To find the optimal value of k
D) To partition data into k distinct clusters based on similarity

9. Which of the following statements best describes an advantage of machine learning in healthcare applications?
- A) Machine learning can replace human doctors and healthcare professionals
 - B) Machine learning cannot handle medical data due to its complexity
 - C) Machine learning can assist in early disease detection and personalized treatment recommendations
 - D) Machine learning techniques are too expensive to implement in healthcare settings
10. What role does an SLA play for CSPs in cloud computing?
- A) It defines the customer's service requirements
 - B) It specifies the maximum amount of data a customer can store
 - C) It helps CSPs avoid responsibility for service inefficiencies
 - D) It provides a detailed overview of cloud security measures
11. Which service model provides access to various software applications over the Internet without requiring users to install the software on their local machines?
- A) Platform-as-a-Service (PaaS)
 - B) Infrastructure-as-a-Service (IaaS)
 - C) Software-as-a-Service (SaaS)
 - D) Community cloud
12. What does the naive approach for offloading data in IoT deployments rely on for decision making?
- A) Historical data
 - B) Machine learning
 - C) Statistical measures
 - D) QoS optimization
13. How can a properly designed IoT architecture impact network bandwidth and energy consumption?
- A) It increases network bandwidth and conserves energy
 - B) It decreases network bandwidth and consumes more energy
 - C) It has no effect on network bandwidth or energy consumption
 - D) It depends on the specific IoT application
14. How are hosts in a network separated from each other?
- A) By different operating systems
 - B) By unique device identifiers like IP addresses and MAC addresses
 - C) By physical distance
 - D) By data encryption
15. Which type of database management systems are commonly associated with structured data?
- A) NoSQL databases
 - B) Object-oriented databases
 - C) Relational database management systems (RDBMS)
 - D) Graph databases
16. How do digital sensors typically represent the output of a quantity being measured?
- A) In the form of binary output signals (logic 1 or logic 0)
 - B) In the form of analog voltage signals
 - C) In the form of continuous waveforms
 - D) In the form of serial transmission
17. What is a significant advantage of a mesh network topology?
- A) Low installation cost
 - B) Simplicity in forming physical connections
 - C) Robustness and resilience in case of link failures
 - D) High data load on a single host
18. What is the origin of the word "topology" in the context of computer networks?
- A) Latin word meaning "communication"
 - B) Greek words "topos" meaning place and "logos" meaning study
 - C) Sanskrit word for "network"
 - D) French word for "connection"
19. In which types of networks are point-to-multipoint connections commonly used?
- A) Wired networks and satellite communication
 - B) Fiber-optic networks and mesh networks
 - C) Wireless networks and IP telephony
 - D) Point-to-point networks and LANs
20. What is the primary purpose of point-to-point connections in networking?
- A) To connect multiple hosts in a network
 - B) To establish direct/dedicated connections between two hosts
 - C) To create a decentralized network topology
 - D) To enable wireless communication

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1. a)	Describe the IoT Networking Components.	8	L*2	1	1
b)	What are the key factors that influence the selection of sensors for IoT sensor nodes in sensing solutions? Explain.	8	L2	2	1
2. a)	What is the ISO-OSI model, and how does it organize networked communication devices into seven distinct layers of abstraction?	8	L1	1	1
b)	What are the key characteristics and attributes that distinguish the various types of actuators?	8	L2	2	1
3. a)	What is Networking? Explain the categories based on network reachability.	8	L2	1	1
b)	What are the different types of sensorial deviations that are typically categorized as errors in sensors. Explain.	8	L2	2	1
Unit – II					
4. a)	What factors are to be considered while deciding on the data offload location? Explain.	8	L2	3	1
b)	Describe the components of an agricultural IoT.	8	L2	4	1
5. a)	Depending on the urgency of data processing, how are IoT data classified? Explain.	8	L2	3	1
b)	Design a case study on Smart irrigation management system.	8	L2	4	1
6. a)	What are the critical factors to be considered during the design of IoT devices? Explain.	8	L2	3	1
b)	What are the key characteristics and attributes that distinguish the various types of actuators?	8	L2	4	1
Unit – III					
7. a)	What are the potential benefits and advantages of implementing Internet of Things (IoT) technology in the healthcare industry, and how can these advantages positively impact patient outcomes and healthcare providers?	8	L2	5	1
b)	What is k- nearest neighbour (KNN)? Explain the steps involved in the implementation of KNN.	8	L2	5	1
8. a)	Explain the concept of decision tree along with an example.	8	L2	5	1
b)	Explain the architecture of healthcare IoT.	8	L2	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
